AWS Labs 4-6

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Purpose:

The purpose of doing these labs was to familiarize ourselves with these new concepts that were talked about in the lectures that we watch. They are to prepare us for the upcoming AWS cloud foundations certification that we are going to take. The labs are also continuing to familiarize us with the AWS UI.

Background:

AWS EBS is a service provided by AWS that provides high performance block storage that is scalable. EBS can be used with EC2 instances to put instances on the storage blocks. EBS volumes can be attached to the EC2 instances to provide that storage for EC2. The volumes are basically hard drives on a regular computer. If you want to back up EBS volumes, you can use EBS snapshots. You can restore the volumes that you backed up anytime you need them. EBS provides many different volumes that you can use to optimize your performance and cost efficiency. One strength of EBS and AWS is that you are able to pay for a volume that suits your current needs. If you need to allocate more resources, you can upgrade your volume easily with EBS. When you buy your servers and hardware, you have to plan for the future which doesn't always maximize profits. Future changes in hardware usage cannot always be predicted. AWS and EBS allow you to change the amount of resources that you use on the spot without extra up-front costs.

An AWS database provides storage where you can store and retrieve data. AWS databases are designed to be reliable, secure, and scalable. AWS databases also follow the p ay as you go model which is much more cost effective for many users whose amount of resources used vary. This is opposed to the traditional way of buying your own servers for your company and allows you to not have to future proof which lowers profit. AWS manages the databases for you, so you are not the one managing it. This makes managing a database much simpler because you are not the one who is managing the database.

Amazon Relational Database Service (Amazon RDS) is the service that allows you to set up a database. It allows you to manage things like storage and cost, but has administrators from AWS do the time-consuming tasks. This allows you to focus on your business.

Elastic load balancing works with EC2 to distribute traffic among different EC2 instances. Elastic load balancing will automatically provide the required amount of load balancing needed to router the traffic. It also provides fault tolerance and redundancy because if one of the instances or databases goes down, the traffic is load balanced on multiple different databases and instances and Elastic load balancing can move traffic to the servers that are still up.

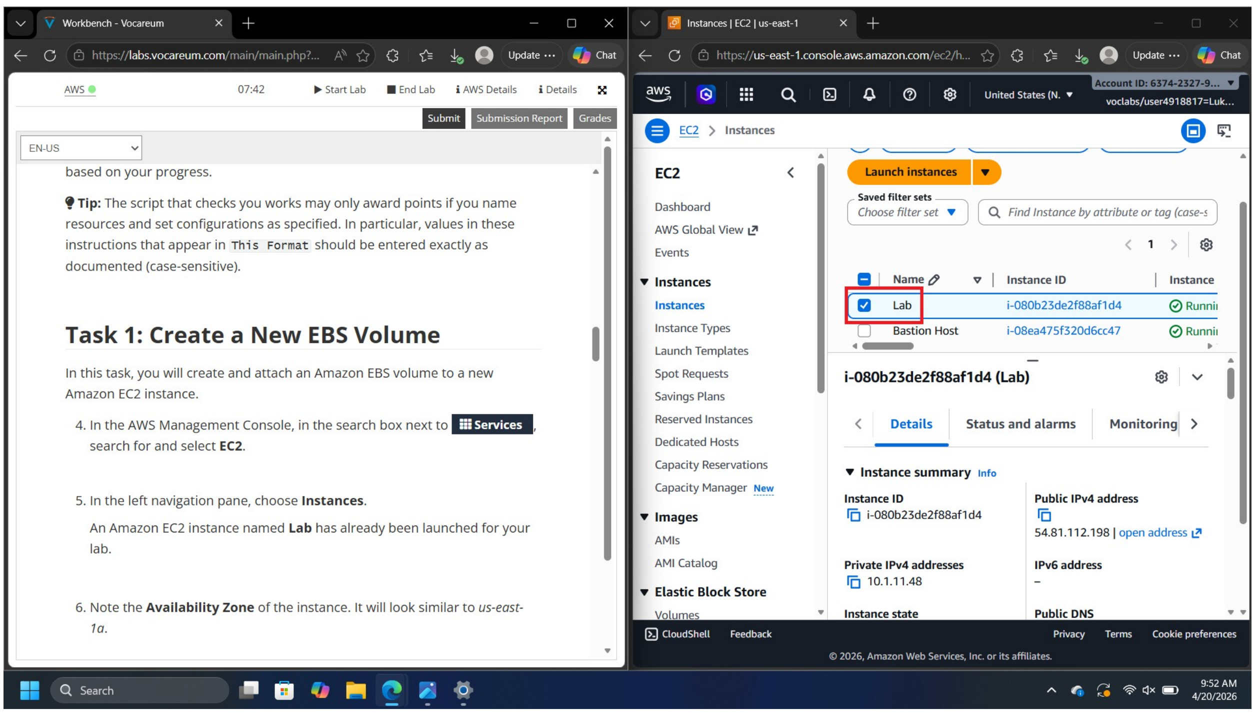
Auto scaling helps you keep applications available. Auto scaling also allows you to scale AWS EC2 capacity up or down based on what you have defined and what you want. You can use auto scaling to verify that are actually running the desired number of instances as sometimes what is automatically configured does not exactly fit what is desired. Auto scaling importantly scales based on the demand at the time. If there is a high demand for resources, auto scaling will automatically increase the bandwidth when needed. While auto scaling is fit for frequent variation, it is also fit for stable demand.

Lab Summary

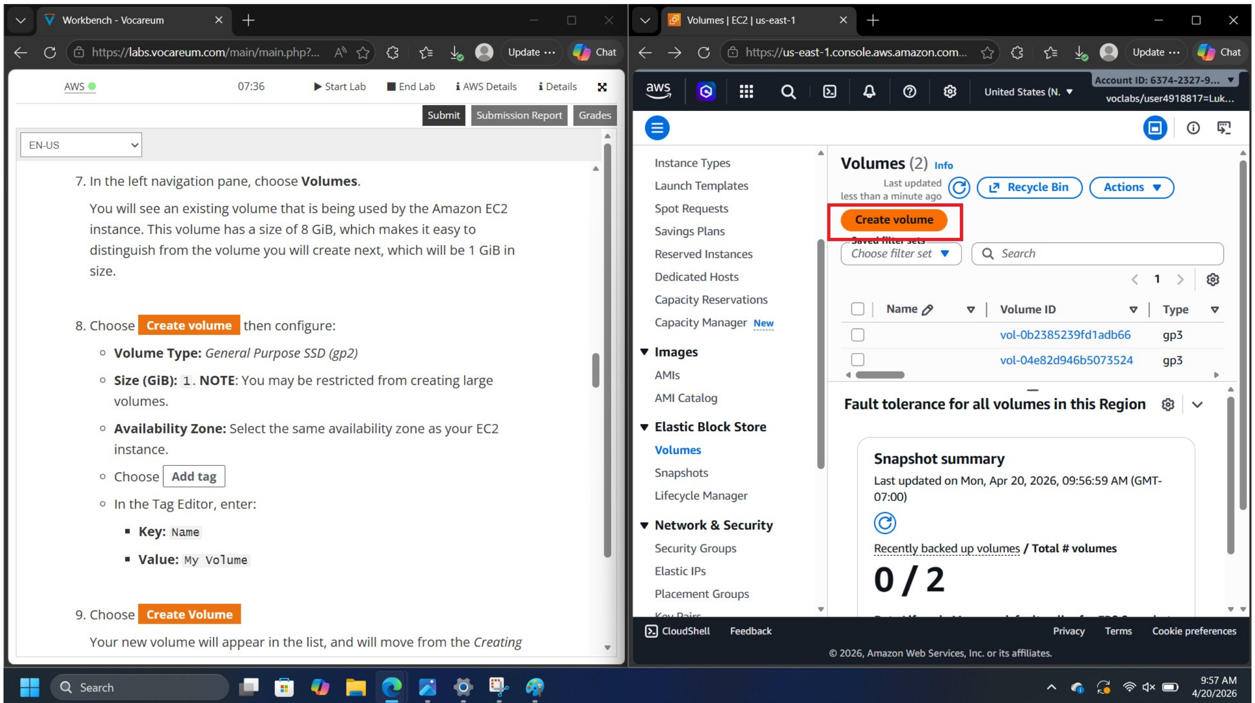
We did three more labs using AWS. We learned about 3 more services that are used with EC2 which are: EBS, RDS, and auto scaling. These are services that provide customizability for EC2 storage, storage that is managed by AWS and not yourself, and automatic scaling of storage when it is needed or desired.

Configs:

Start the lab as usual and open EC2



Choose instances there should be an instance named lab like shown above



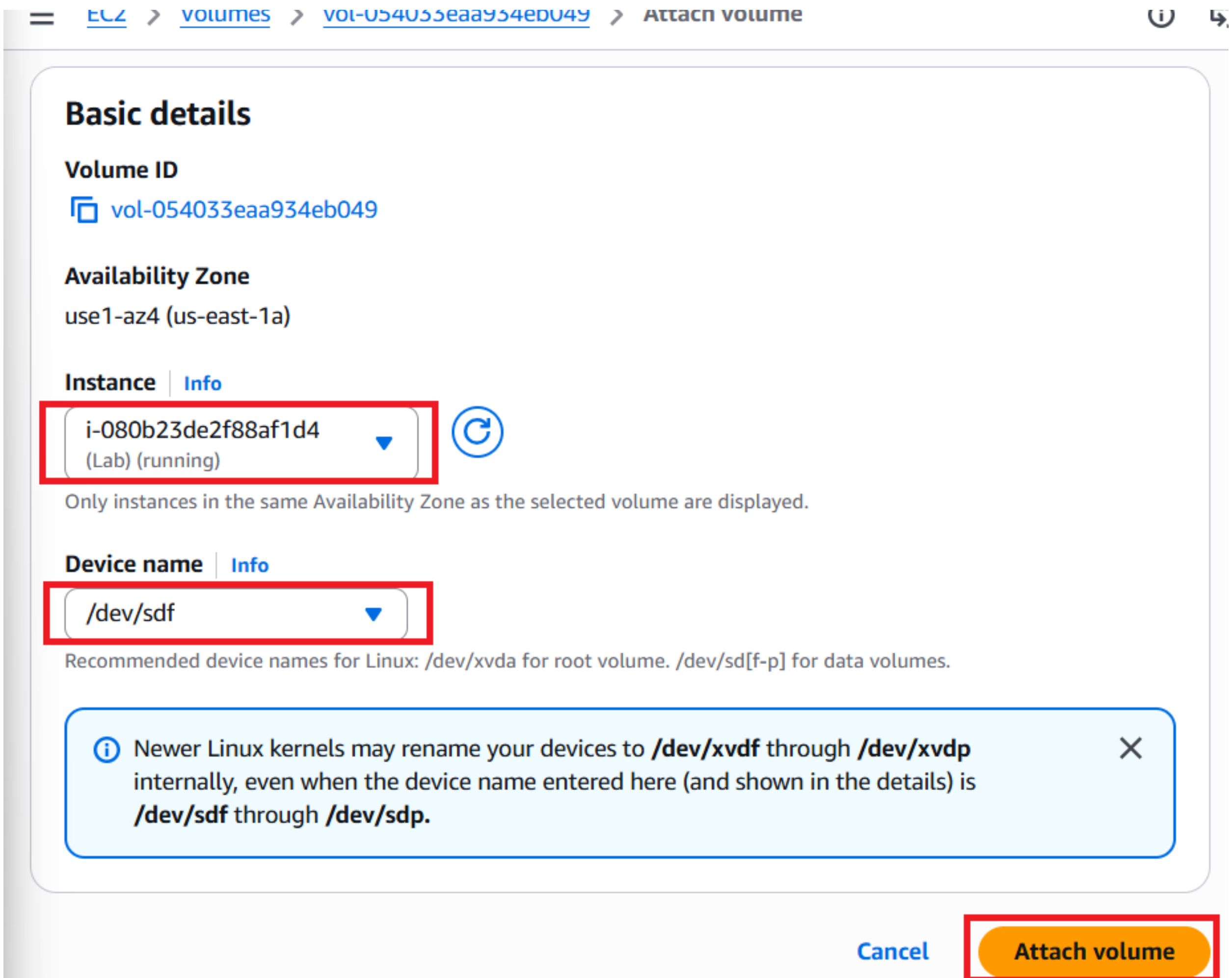
Now choose create volume and configure it with the following configs:

> **"Volume Type: *General Purpose SSD (gp2)***
> **Size (GiB): 1. NOTE: You may be restricted from creating large volumes.**
> **Availability Zone: Select the same availability zone as your EC2 instance.**
> **Choose Add Tag**
> **In the Tag Editor, enter:**
> > **Key: Name**
> > **Value: My Volume" (AWS Academy)**

Choose create again

Select my volume

Find the actions menu and choose attach volume

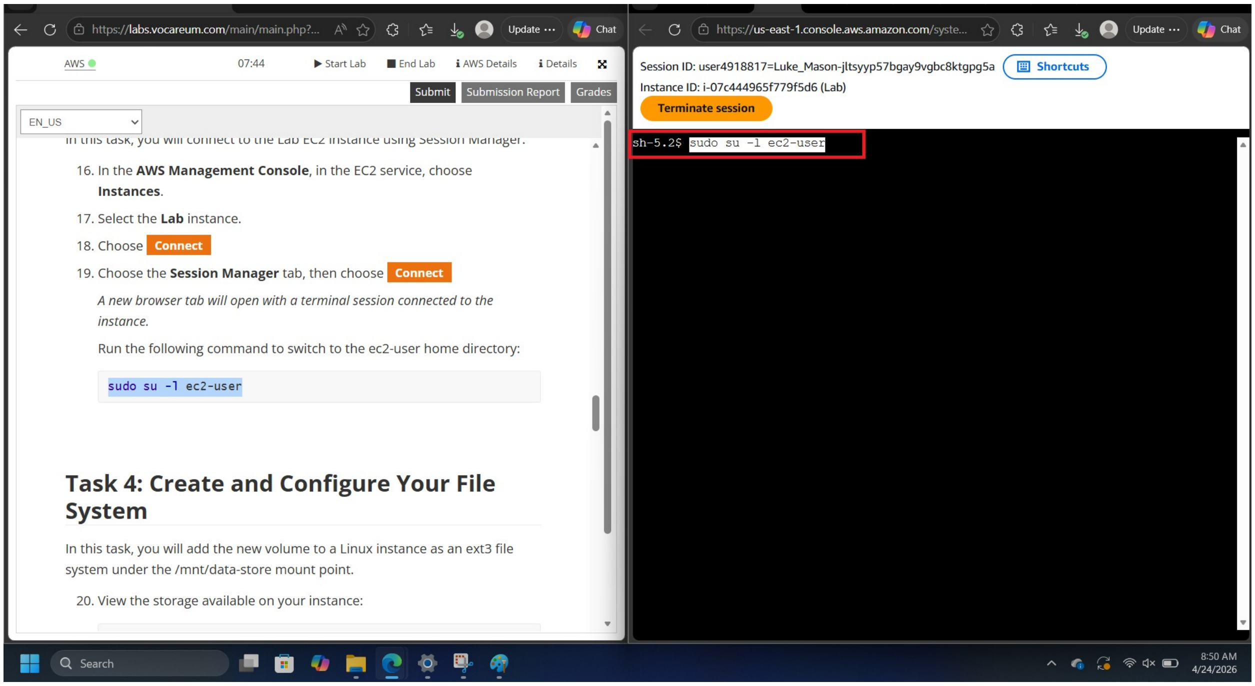


In the instance field select the lab instance

Configure the basic details like the screenshot above.

Find instances and choose the lab instance and click connect. Find the session manager and choose connect again.

There should be a new tab with a terminal that opened keep this tab open.



Use the command: `sudo su -l ec2-user`

To view the storage for your instance do df –h

You will not see the new volume that you have created yet.

Create the ext3 file system: `sudo mkfs -t ext3 /dev/sdb`

Create a directory: `sudo mkdir /mnt/data-store`

Mount the new volume: `sudo mount /dev/sdb /mnt/data-store` (this will mount the volume every time the instance is started: `echo "/dev/sdb   /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab`)

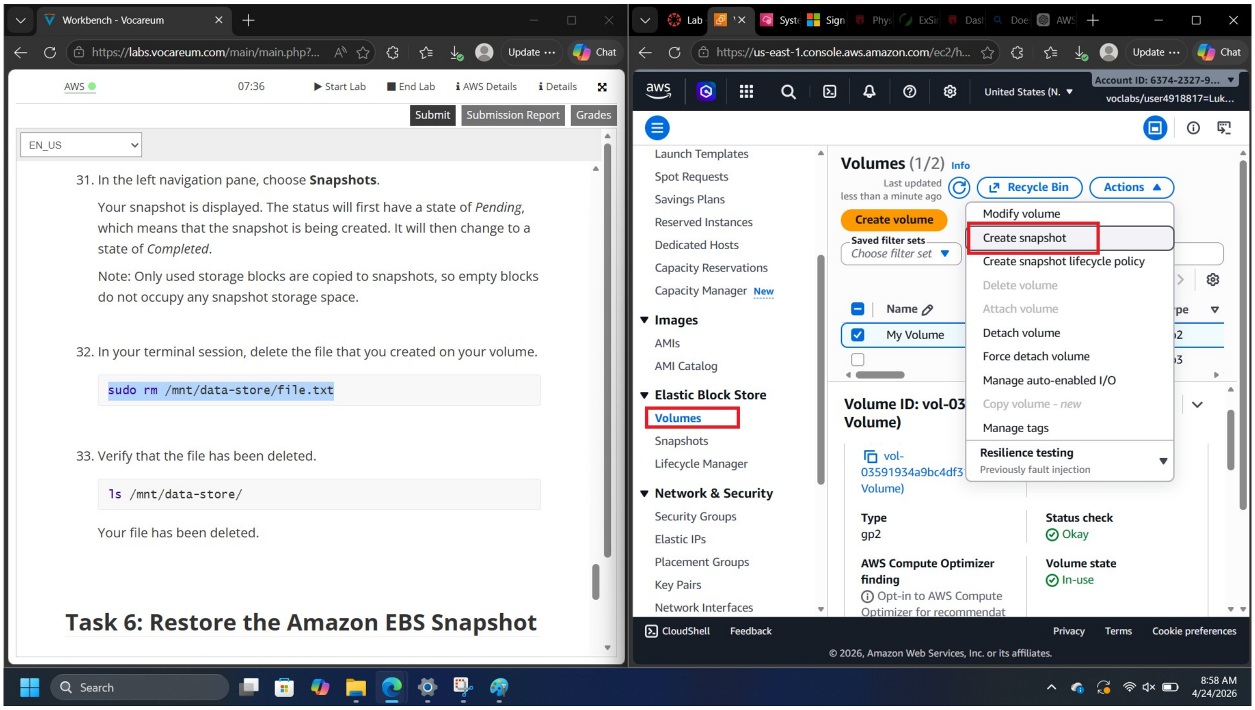
View the last setting: `cat /etc/fstab`

View the storage again with df –h

There will now be an additional line

```
sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"
cat /mnt/data-store/file.txt
```

This writes text to your instance.



Choose volumes and then select my volume.

Click on the drop-down actions menu and find create screenshot.

Choose add tag and then configure the key as Name and the value as My Snapshot.

Choose create snapshot.

On the left panel choose Snapshots

You can also delete a snapshot by entering this command into the terminal: `sudo rm /mnt/data-store/file.txt`

Verify that the file has been deleted: `ls /mnt/data-store/`

Select my snapshot then select create volume from snapshot.

Use the same availability zone.

Choose add tag and configure the key to be Name and Value to be Restored volume.

Create the volume

Choose volumes and select Restored Volume.

In the actions drop down menu, choose attach volume.

In the instance field choose lab and for the device name choose /dev/sdc

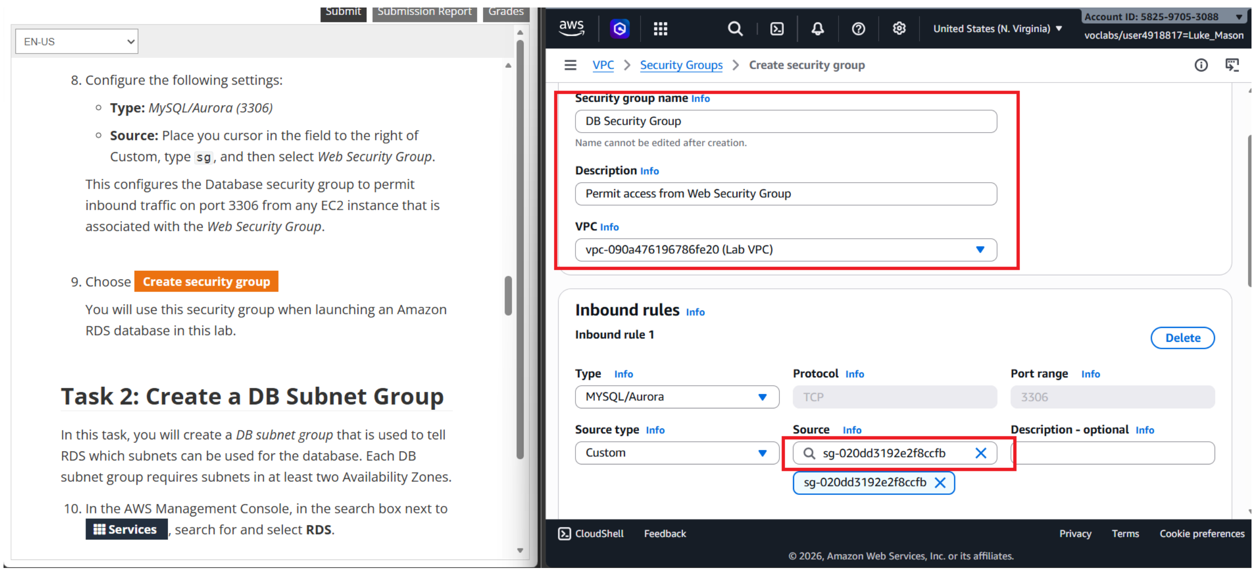
Now choose attach volume.

Create a directory for mounting the new storage volume: sudo mkdir /mnt/data-store2
Mount the new volume: sudo mount /dev/sdc /mnt/data-store2
Verify that volume you mounted has the file that you created earlier. ls /mnt/data-store2/
You should see file.txt.

Amazon RDS lab

Search for Amazon VPC

On the left search for security groups.



Choose create security group and configure the following settings:

> "Security group name: DB Security Group
> Description: Permit access from Web Security Group
> VPC: Lab VPC" (AWS Academy)

In the inbound rules choose add rule:

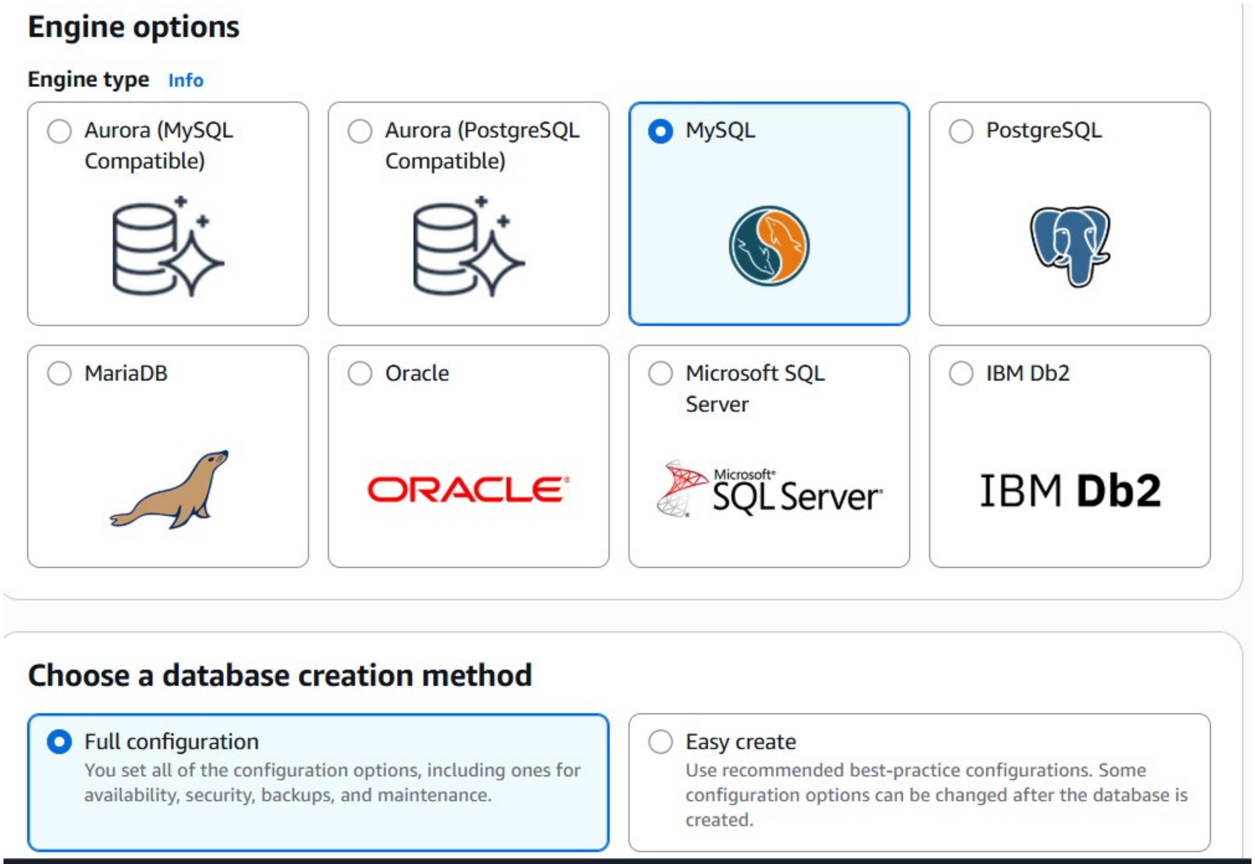
> "Type: MySQL/Aurora (3306)
> Source: type sg and select the Web Security Group." (AWS Academy)

Choose create security group

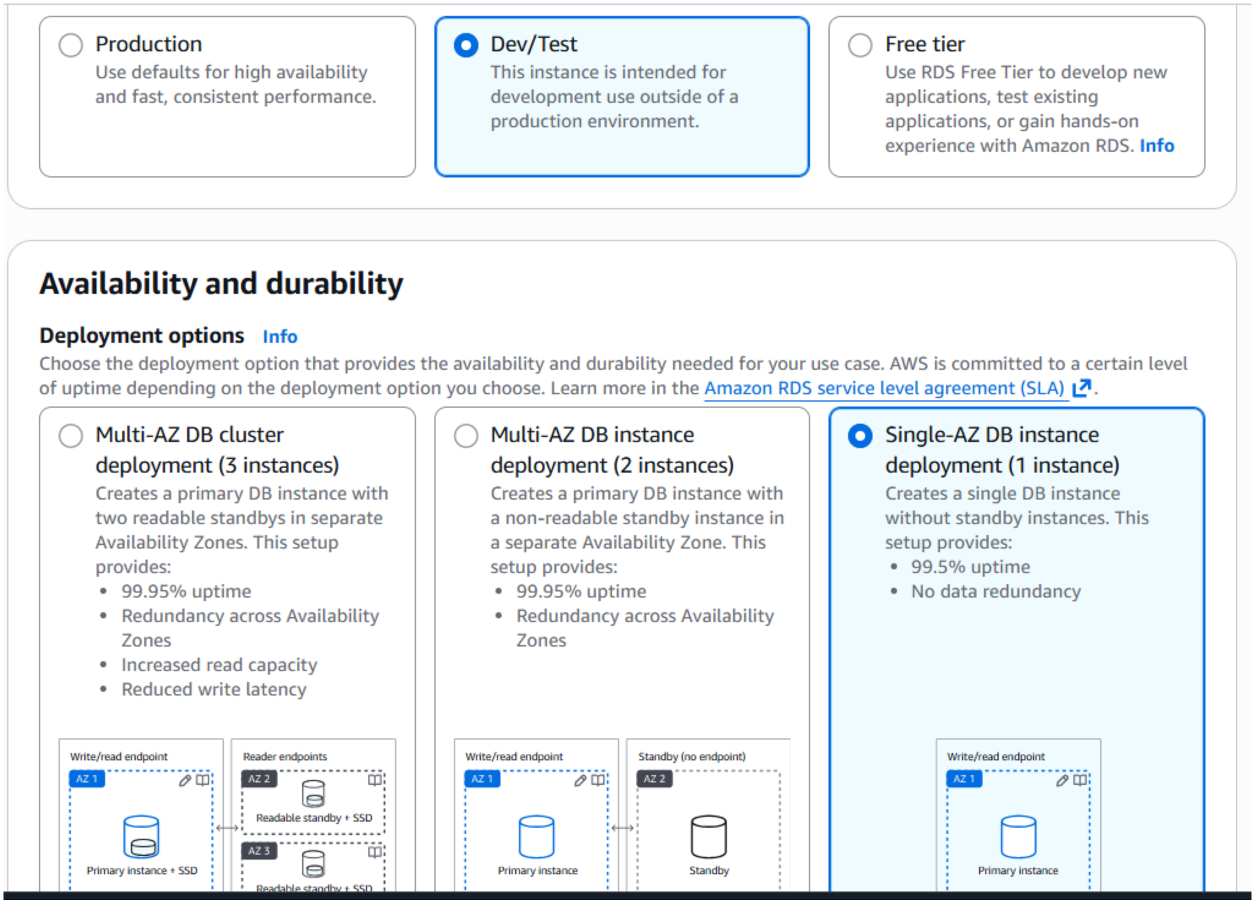
Search for RDS in the search bar at the top and select it
On the left choose subnet groups
Choose Create DB Subnet Group and configure the following:

> "Name: DB-Subnet-Group
> Description: DB Subnet Group
> VPC: Lab VPC" (AWS Academy)

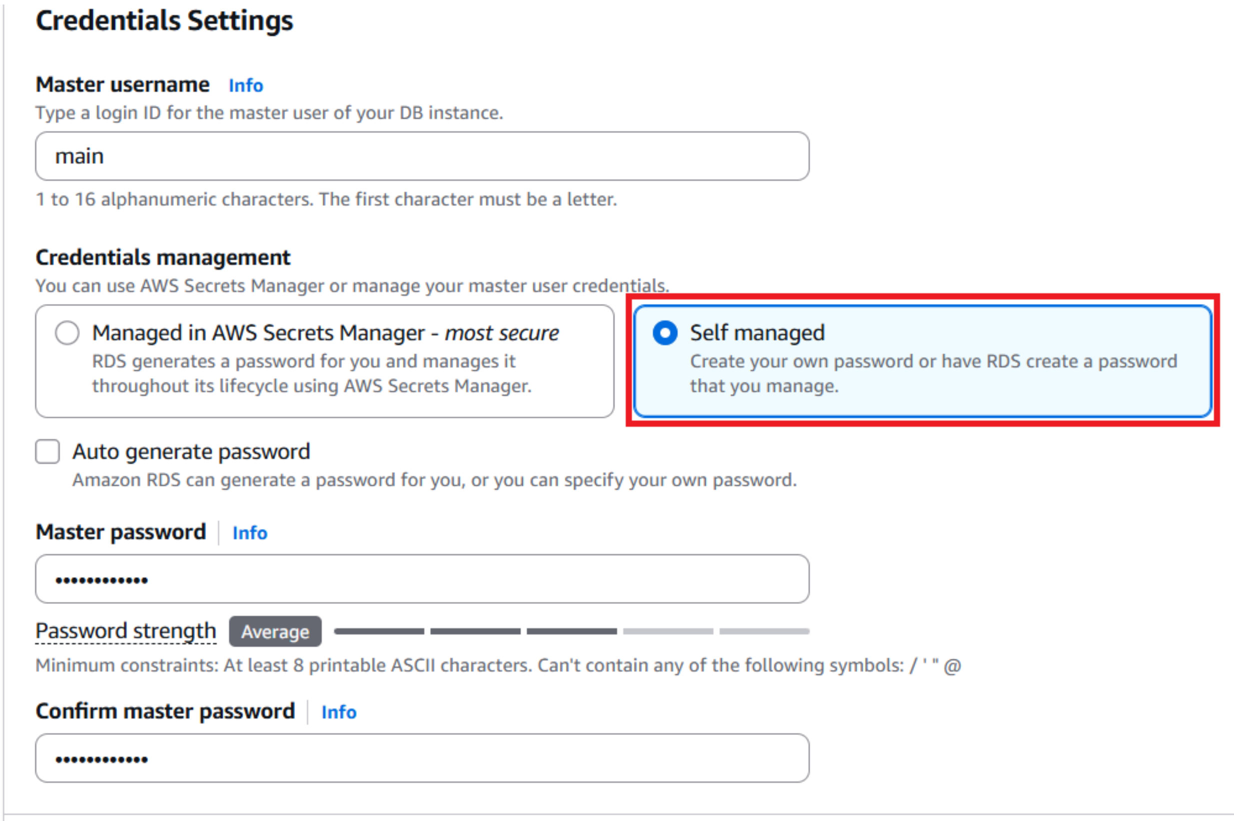
Scroll down to the add subnets
Show all the availability zones and select US 1a and US 1b
Show all the subnets and pick 10.0.1.0/24 and 10.0.3.0/24
Choose create.



Choose databases
Select MySQL
Under the templates tab choose Dev/Test

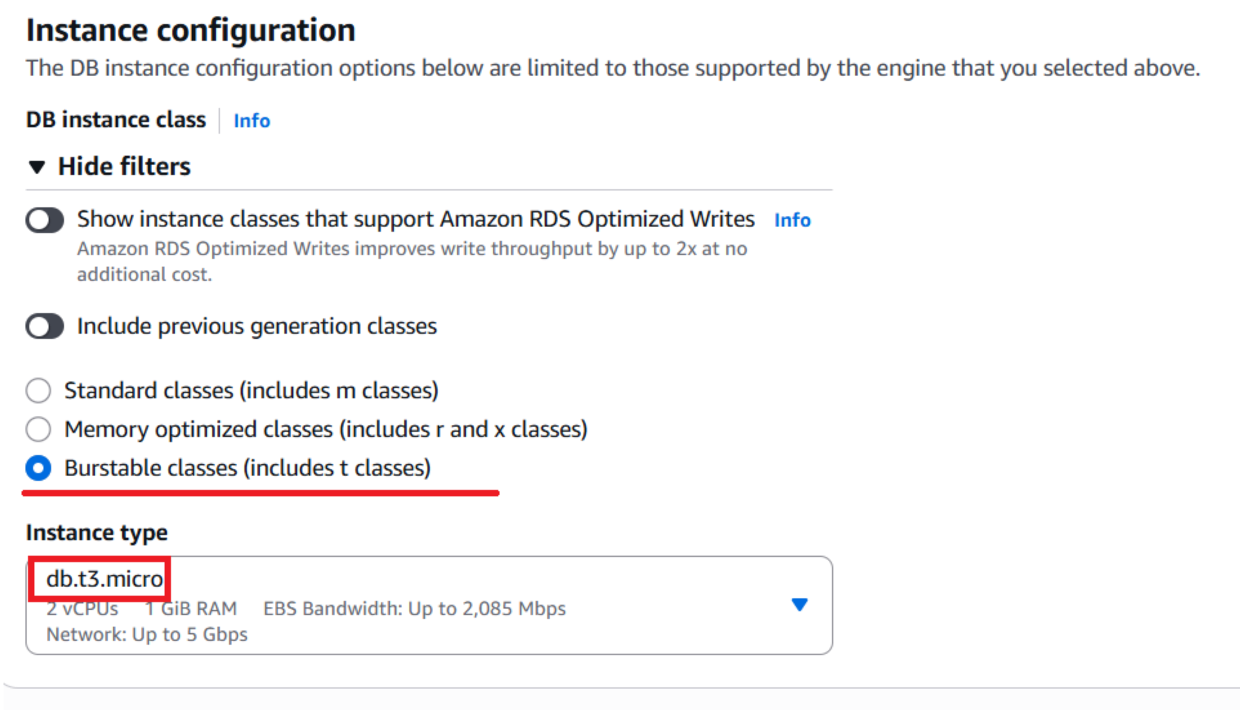


Under Availability and durability choose Multi-AZ DB instance.



In Settings, configure these:

> “DB instance identifier: lab-db
> Master username: main
> Master password: lab-password
> Confirm password: lab-password" (AWS Academy)



Under DB instance class, configure:

“Select Burstable classes.
Select db.t3.micro.”

Under the storage tab, configure:

“Storage type: General Purpose (SSD)
Allocated storage: 20”

Configure Virtual Private Cloud (VPC) to be lab VPC under connectivity.

Under Existing VPC security groups, from the dropdown list:

“Choose DB Security Group.
Deselect default.”

Under Monitoring expand Additional configuration.

“Uncheck Enable Enhanced monitoring”

Under Additional configuration, configure:

“Initial database name: lab
Uncheck Enable automatic backups.
Uncheck Enable encryption” (AWS Academy)

This is the last of the configs

Choose Create database.

Choose lab-db. You might need to wait for a bit.

Wait until it becomes available.

Scroll down to the connectivity and security section and copy the Endpoint field.

Paste it somewhere where you can copy it later.

Click on the I AWS Details on the AWS instructions and copy the IP address.

Open a new tab and paste in the ip address.

Choose the RDS link it should be near the top

Configure the following settings:

> "Endpoint: Paste the Endpoint you copied to a text editor earlier
> Database: lab
> Username: main
> Password: lab-password
> Choose Submit" (AWS Academy)

There should be some sort of address book that is on the screen.
You can change things in the address book to test functionality.

Scaling and load balancing:

In the search box next to services search for EC2

Choose instances

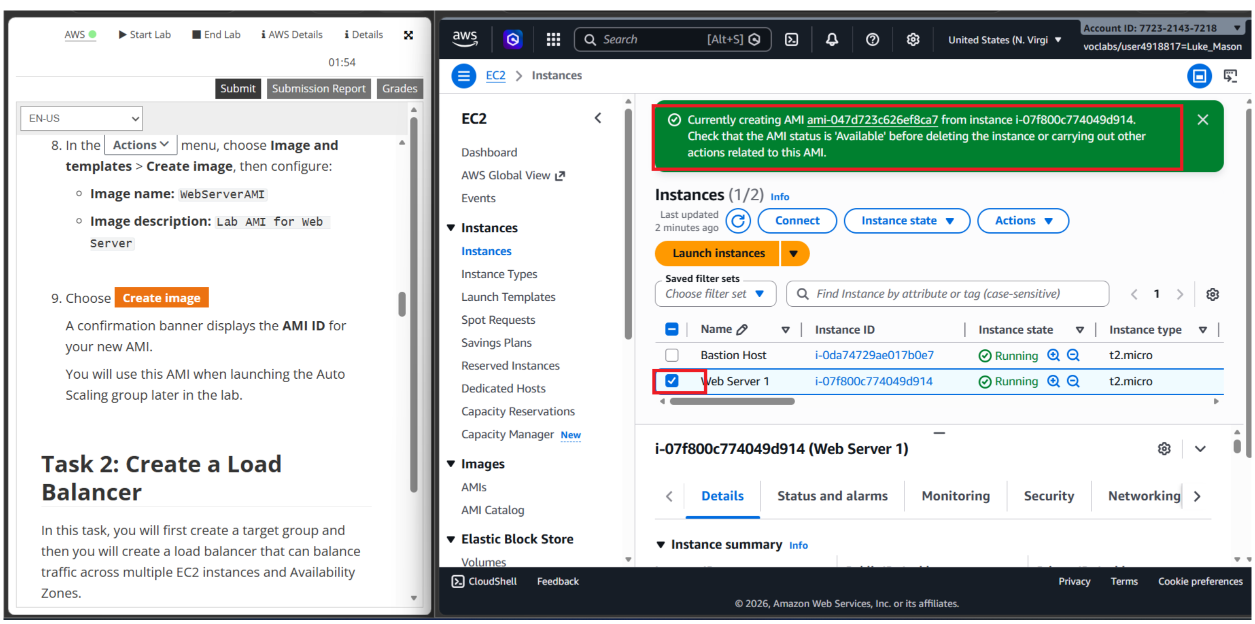
Wait for the instances to fully go up. This could take some time.

Select the Web server 1

In the actions drop down menu choose Images and templates> create image.

Configure the Image name to be WebServerAMI and the description as Lab AMI for Web Server.

Choose Create Image.



"Choose target groups. Configure these settings:

Choose Create Target group.
Choose a target type: Instances
Target group name, enter: LabGroup
Select Lab VPC from the VPC drop-down menu”(AWS Academy)

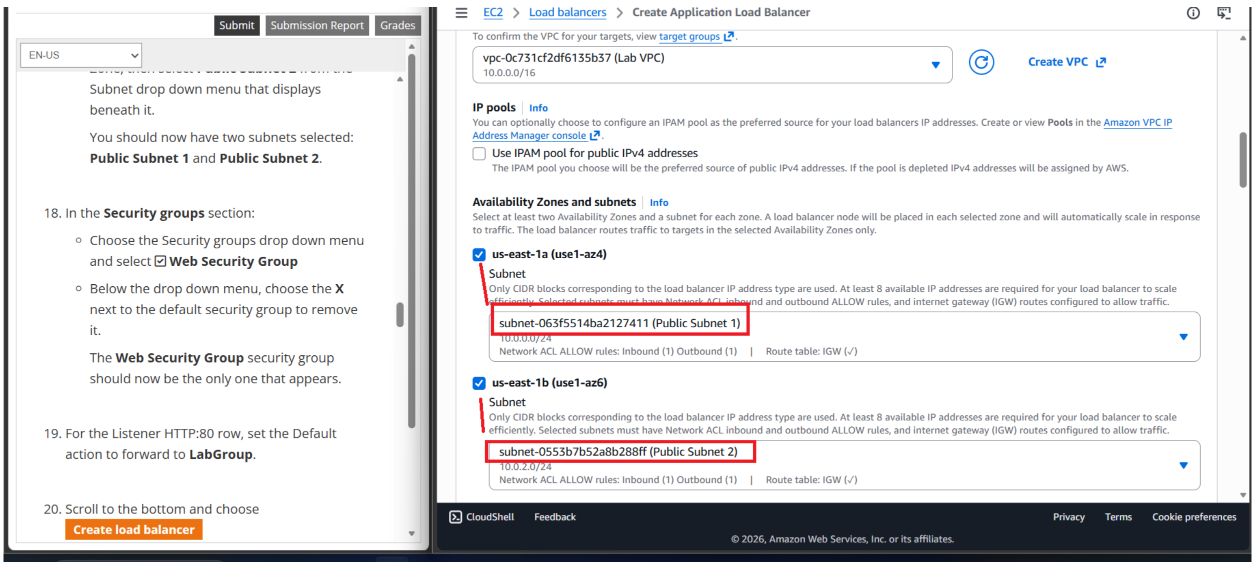
Choose Next and a register target screen appears.

Choose Create Target Group

On the left side of your screen choose Load Balancers then choose create load balancer.

Under the Application load balancer button choose the create button. 4

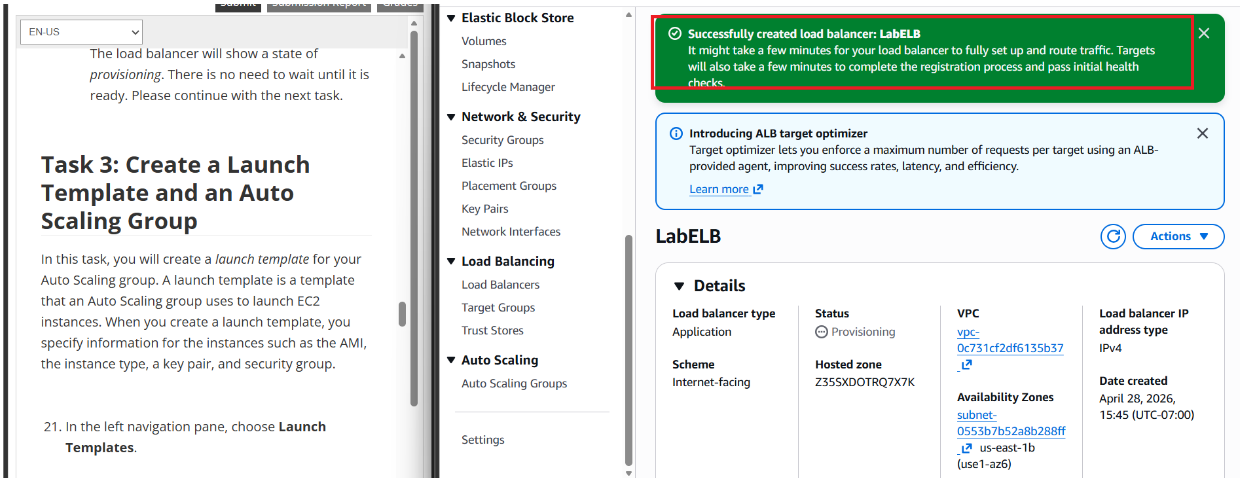
Enter LabELB for the application load balancer



Choose your labVPC for VPC then choose public subnet 1 for the first availability zone and public subnet 2 for the second availability zone.

For the security groups section, select the web security group and remove the default security group.

For the Listener HTTP:80 row, set the Default action to forward to LabGroup

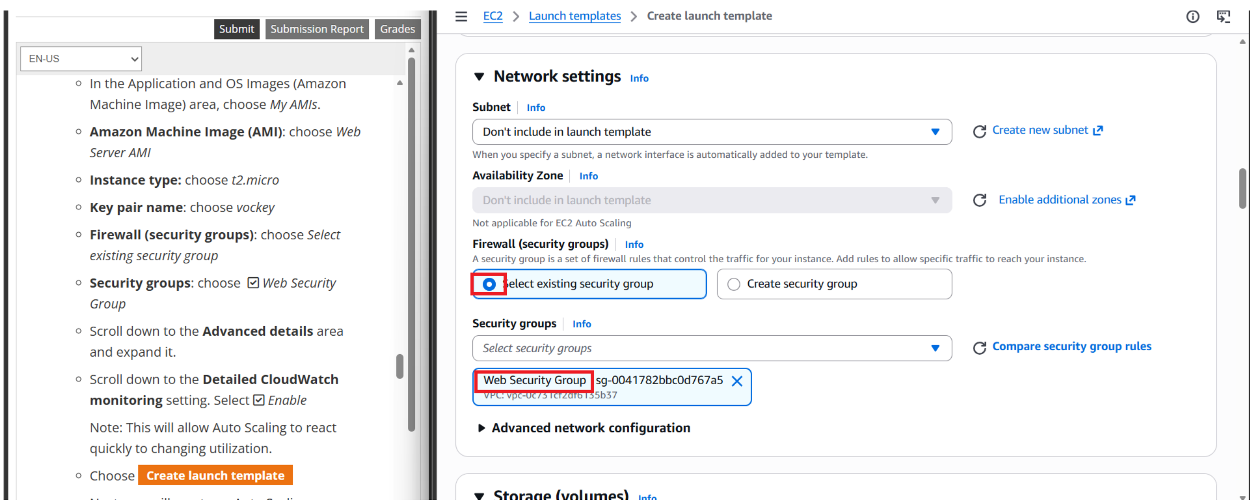


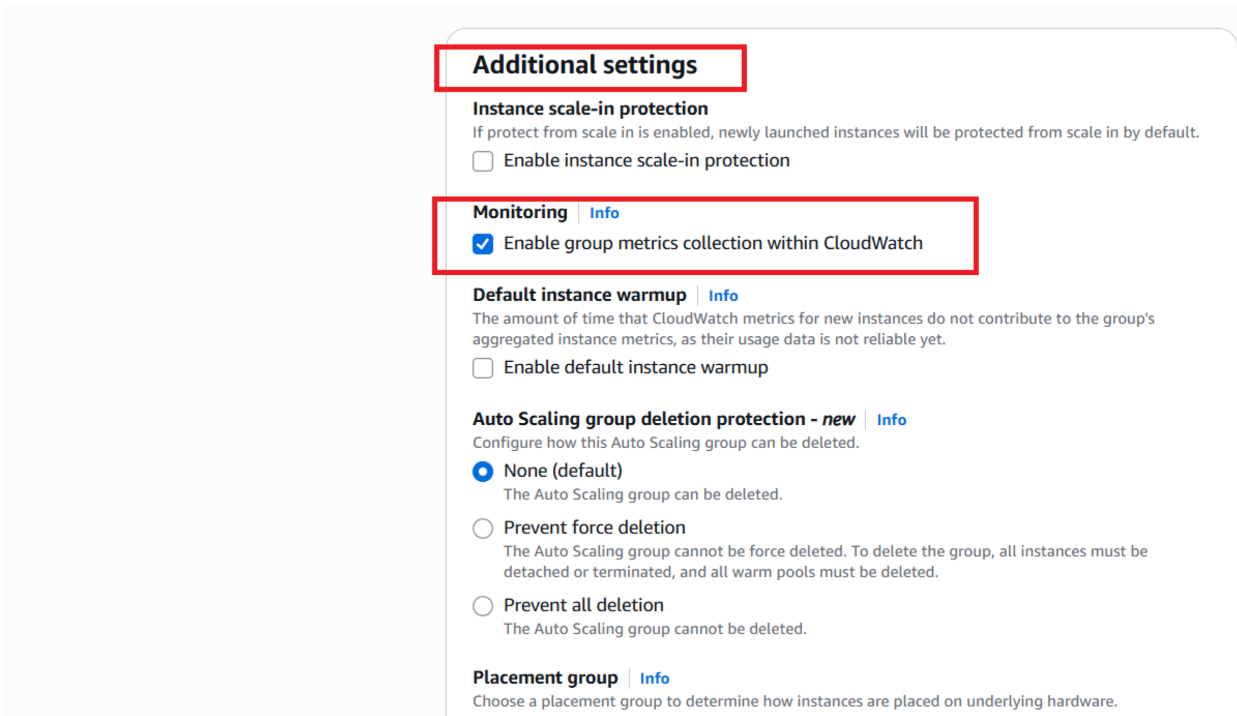
Choose create load balancer

Choose Launch Templates and then choose create launch template.

Configure the following settings:

“Launch template name: LabConfig





Under Auto Scaling guidance, select Provide guidance to help me set up a template that I can use with EC2 Auto Scaling
In the Application and OS Images (Amazon Machine Image) area, choose My AMIs.
Amazon Machine Image (AMI): choose Web Server AMI
Instance type: choose t2.micro
Key pair name: choose vockey
Firewall (security groups): choose Select existing security group
Security groups: choose Web Security Group
Scroll down to the Advanced details area and expand it.
Scroll down to the Detailed CloudWatch monitoring setting. Select Enable" (AWS Academy)
Choose create launch template
Choose the LabConfig template
In the actions drop down menu choose the create auto scaling group.
Configure these: for step 1
"Auto Scaling group name: Lab Auto Scaling Group
Launch template: confirm that the LabConfig template you just created is selected." (AWS Academy)
Choose next
Configure these for step 2:
"VPC: choose Lab VPC
Availability Zones and subnets: Choose Private Subnet 1 and then choose Private Subnet 2" (AWS Academy)
Configure these for step 3"

“Choose Attach to an existing load balancer

Existing load balancer target groups: select LabGroup.

In the Additional settings pane:

Select Enable group metrics collection within CloudWatch

This will capture metrics at 1-minute intervals, which allows Auto Scaling to react quickly to changing usage patterns”

Configure these for step 4:

“Under Group size, configure:

Desired capacity: 2

Minimum capacity: 2

Maximum capacity: 6

This will allow Auto Scaling to automatically add/remove instances, always keeping between 2 and 6 instances running.

Under Scaling policies, choose Target tracking scaling policy and configure:

Scaling policy name: LabScalingPolicy

Metric type: Average CPU Utilization

Target value: 60”

For step 5 keep everything default

For step 6 configure these:

“Choose Add tag and Configure the following:

- Key: Name
- Value: Lab Instance

Choose Next” (AWS Academy)

Review everything and if it looks right then create the auto scaling group.

Choose instances (Auto scaling should have launched 2 instances)

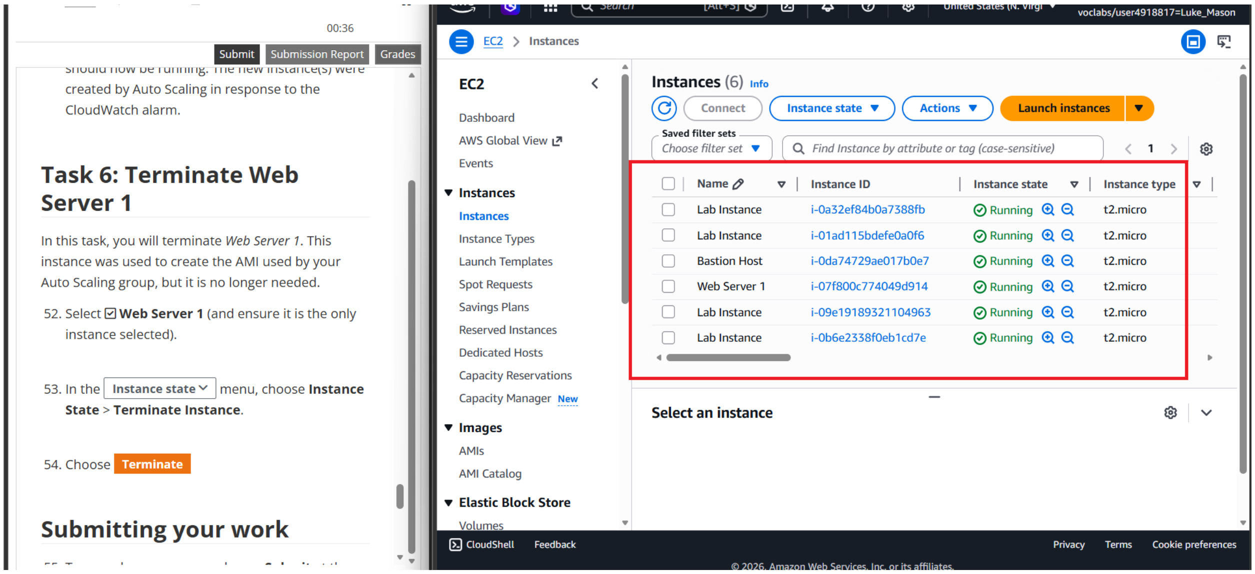
Choose target groups and select LabGroup

Click on the target tab and wait until it says that the status of the instances is healthy. Refresh occasionally. Once the instances say healthy they have passed the test.

On the left choose load balancers

Select LabELB

Go to the details and find the DNS name. Copy it and then paste the DNS name into a browser. If the app appears in the browser you know that load balancing worked.



Search for CloudWatch in AWS and choose all alarms on the left. 2 alarms should be shown. For troubleshooting, check out the AWS Academy instructions. Choose the alarm that says Alarmhigh in its name and refresh if the alarms do not say OK.

Navigate back to the browser and select the load test next to the AWS logo.

If you return to CloudWatch you will see that the alarm high will be in alarm.

If you go to EC2 and go to instances you can see that new instances have been create to account for the increased traffic.

Select WebServer1 and go to instance state>Terminate instance to terminate the instance

Problems:

The AWS labs are slightly different from the regular labs that we do. They provide a little more instruction to us, which decreases the amount of issues and troubleshooting that we have to do as a group. I also had issues with trying to locate some settings that were either in the wrong spot or incorrectly documented. This greatly increased the amount of time that the labs took because I would have to restart the labs if they were not completed in under two hours. I had to restart specifically the second and third lab multiple times.

Conclusion

I think that these labs helped give me a baseline and introduction for AWS and could computing. The labs helped show me the AWS console and several important services that would be commonly used. While I do think that they did teach some of the basics of the AWS console, I think that they could have been better by excluding a lot of steps that were unnecessary and time consuming.